

Technology Stack

Date	26 June 2026
Team ID	-
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, JavaScript (ES6)	Lightweight, dependency-free UI that loads fast and keeps the app easy to deploy on static hosts like Vercel/Netlify.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Fast API (Python)	Fully asynchronous, high-performance framework with automatic Swagger/ReDoc API docs.
3	Authentication	JWT (python-jose) + bcrypt/passlib	Stateless, secure session handling and safe password storage
4	Database	MongoDB Atlas + Motor (async driver)	Flexible document storage suited to varied AI-response payloads; async driver avoids blocking the event loop.
5	AI / LLM Engine	Google Gemini API (google-generative AI)	Powers Q&A, explanations, quiz/roadmap JSON generation, and summarization with high-quality responses.
6	API Layer	python-multipart, HTTPS, slow API	Handles file uploads, outbound HTTP calls, and rate limiting.
7	Validation	Pydantic / pydantic-settings	Type-safe request/response models and centralized configuration management.

8	Deployment	Render / Railway / Fly.io (backend), Vercel / Netlify (frontend	Simple, low-cost hosting options suited to a Fast API + static frontend split.
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